CIST 3222 DATABASE SYSTEMS
GROUP PROJECT REPORTS

Boardwalk Casino Database Project

Group 3
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Project Overview

1. Executive Summary

Our business, Boardwalk Casino, was constructed in order to bring a friendly-family game experience to the people of Sea Isle City. The goal of Boardwalk Casino is to have customers enjoy their time playing different games and redeeming tickets for prizes to create and take memories home after visiting. However, Boardwalk Casino faces many issues such as tracking inventory, analyzing customer interests, maintenance tracking. The current methods of tracking these data points were either done manually with pen and paper or visually seeing popularity with games.

To better handle these issues, it was decided that a database system was necessary. The new database system is able to keep track of customers, game analytics, game maintenance logging, vendors, orders, and inventory levels. This would lead to properly ordering correct amounts of product to keep redemption prizes in stock and still keep a profit.

Using this system, Boardwalk Casino would be able to efficiently keep track of what products are being played the most, what prizes are being redeemed the most, and what vendors offer the lowest price. Using this database allows the owner to free up time to focus energy into other parts of the business.

2. Team Contribution

Daniel	Business Description, Data Model, Entity-Relationship Diagram, Create/Insert Scripts
Christian	Business Description, Data Model,Entity-Relationship Diagram, Create/Insert Scripts
Sean	Business Description, Entity-Relationship Diagram, Create/Insert Scripts
Gavin	Data Model, Entity-Relationship Diagram, Presentation. Create/Insert Scripts
Josh	Business Description, Presentation

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1. Business Description

1.1 Name of Business

Boardwalk Casino

1.2 Purpose of Business

Boardwalk Casino is an arcade and entertainment business located on the boardwalk in Sea Isle City. The company's purpose is to provide visitors with engaging games, ticket-based prize redemption, and a family-friendly experience that encourages repeat seasonal customers.

A well-structured database is essential for organizing daily operations—managing customer data, tracking inventory, and analyzing performance metrics—to support data-driven decision-making and operational efficiency.

1.3 Summary of Business Activities

Boardwalk Casino's core activities center around two main areas:

- Games Operations: Maintaining, repairing, and purchasing arcade machines; monitoring plays, payouts, and performance metrics.

- Prize and Inventory Management: Ordering new prizes, tracking stock levels, managing vendor relationships, and ensuring that popular prizes remain available throughout the season.

These processes require continuous monitoring of transactions, inventory quantities, and customer engagement—functions that a centralized database can automate and streamline.

1.4 Problems, Opportunities, and Objectives

Current Problems

- Inconsistent and inaccurate inventory tracking leads to over- or under-stocking prizes.

- Manual methods for analyzing popular games and prizes cause delays in operational decisions.
- Revenue and performance calculations are performed manually, reducing efficiency.

Opportunities

- Automate data collection from game machines and prize counters.
- Use analytics to identify high-performing games and high-value customers.
- Improve vendor coordination and restocking decisions based on real-time data.

Objectives

- Maintain accurate, real-time inventory across crane and redemption systems.
- Track and analyze customer behavior to improve seasonal offerings.
- Generate automated financial and performance reports to guide management decisions.

1.5 Business Case

The database will address several operational challenges and provide measurable benefits:

- **Customer Tracking**: A CUSTOMER entity will store information about new and returning visitors, helping identify frequent players and popular games. Seasonal data trends will inform marketing and promotions.
- **Inventory Management**: By tracking both crane and redemption prize inventories, the system will identify fast-moving and slow-moving items. This ensures that popular prizes remain stocked and that capital isn't tied up in unused inventory.
- **Game Analytics**: Storing revenue, play counts, and payout rates will allow management to analyze which games generate the highest profit or require maintenance or replacement.
- **Card Management**: The database will store game-card balances and transaction histories, allowing staff to reissue lost or broken cards and maintain accurate records of credits and bonuses.

Overall, the database will replace manual processes with an integrated system that supports accurate record-keeping, efficient operations, and assist decision making.

1.6 Information and Data Requirements

Data to Be Stored (Raw Data):

- Customer details (name, contact information, date of birth, card balances)
- Game information (machine ID, cost per play, payout rate, type, MAC address)
- Game card transactions (credits, bonus, balance)
- Employee records (ID, name, pay rate)
- Prize inventory details (item name, type, vendor, wholesale cost, retail value, quantity)
- Vendor information (name, phone, email)

Information to Be Generated (Insights):

- Top performing games by revenue and usage
- Popular and low-performing prizes based on redemption frequency
- High-value or frequent customers
- Inventory turnover rates and vendor performance
- Seasonal trends in customer visits and sales

1.7 List of Entities

Each includes its primary key (underlined) and key attributes. Relationships exist among customers, game cards, games, employees, vendors, and inventory items.

CUSTOMER (<u>CustomerID</u>, FirstName, LastName, PhoneNumber, Email, DOB, QuantityOfCards)
GAME_CARD (<u>CardNumber</u>, CardHolderName, CurrentCredits, CurrentBonus, CustomerID)
GAMEPLAY_HISTORY(<u>PlayID</u>, CardNumber, GameID, PlayTime, CreditsUsed, TicketsWon)
GAME (<u>GameID</u>, GameName, PurchasePrice, PlayCost, TypeOfGame, MACAddress)

EMPLOYEE (EmployeeID, FirstName, LastName, PayRate)

VENDOR (VendorID, VendorName, Phone, Email, Website, ProductType)

ORDER(OrderID, OrderDate, OrderTotal, VendorID, EmployeeID)

ORDER_ITEMS(OrderID, ItemID, Quantity, UnitCost)

PRIZE_INVENTORY (ItemID, ItemName, PrizeType, Manufacturer, WholesaleCost, RetailValue, Quantity)

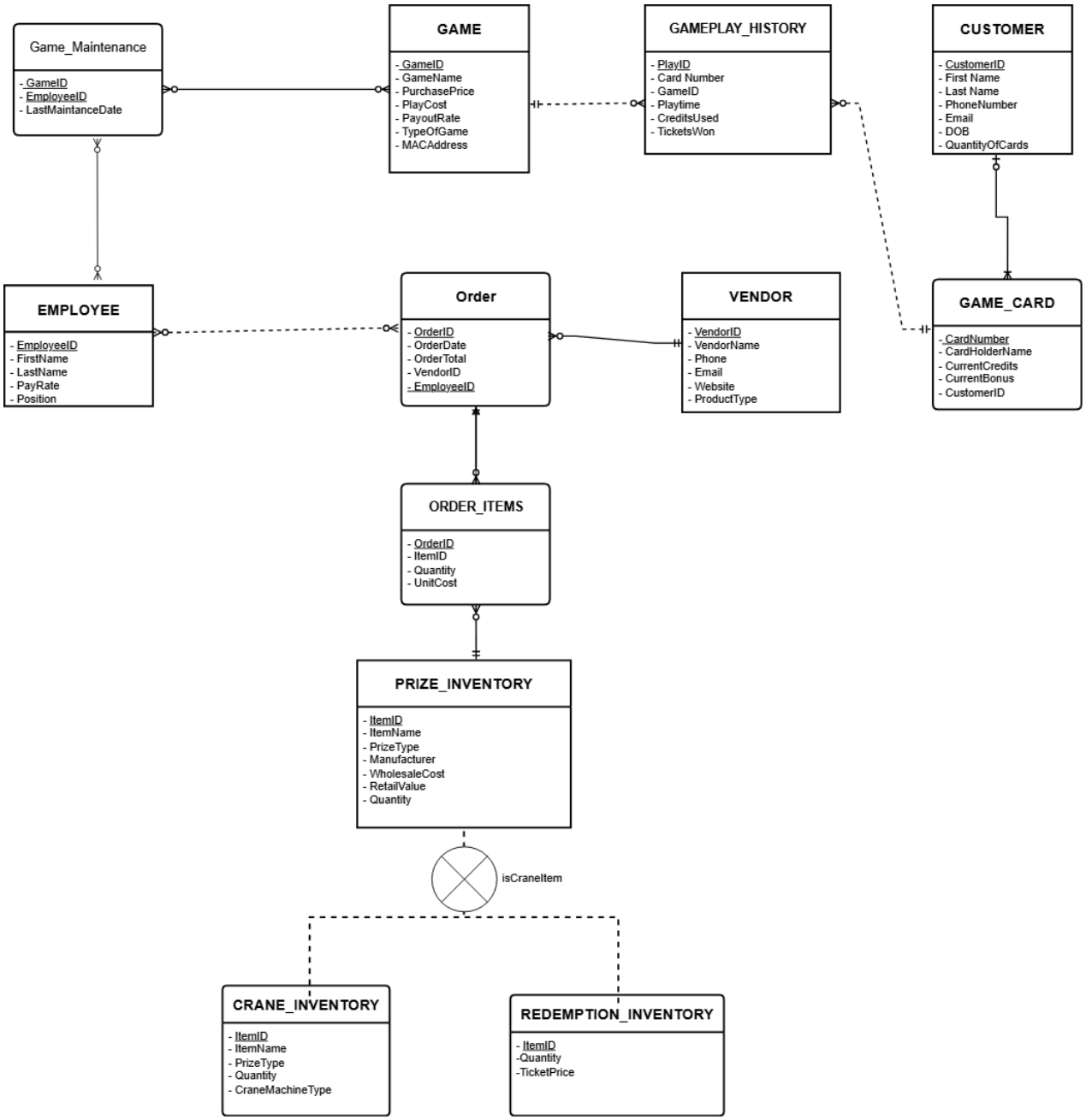
CRANE_INVENTORY (ItemID, ItemName, PrizeType, Quantity)

REDEMPTION_INVENTORY (ItemID, Quantity, TicketPrice)

2. Conceptual Data Model

2.1 Entity-Relationship Diagram

Underlined Attributes are Primary Keys



2.2 Relationship Sentences and Explanations

- **Customer and Game_Card**
  - **Type of relationship**: One-to-many (1:M), optional-to-mandatory

- **Description:** A customer can own zero or many game cards; every game card belongs to exactly one customer.
- **Minimum Cardinality**: A customer may have 0 cards.
- **Maximum Cardinality**: A customer can have many cards.

- **Game and GamePlay_History**
  - **Type of relationship**: One-to-many (1:M), optional-to-mandatory
  - **Description:** A game can appear zero or many times in play history; every play history row records exactly one game.
  - **Minimum Cardinality**: A game may have 0 history rows.
  - **Maximum Cardinality**: A game can have many history rows.

- **Game_Card and GamePlay_History**
  - **Type of relationship**: One-to-many (1:M), optional-to-mandatory
  - **Description:** A game card can appear zero or many times in play history; every play history row records exactly one card.
  - **Minimum Cardinality**: A game card may have 0 history rows.
  - **Maximum Cardinality**: A game card can have many history rows.

- **Game_Card and Game (via GamePlay_History)**
  - **Type of relationship**: Many-to-many (M:N), optional-to-optional, resolved through GAMEPLAY_HISTORY
  - **Description:** A card can play many games, and a game can be played by many cards; each play is captured as one row in GAMEPLAY_HISTORY.
  - **Minimum Cardinality**: A card may play 0 games; a game may be played by 0 cards.
  - **Maximum Cardinality**: Many on both sides.

- **Vendor and Order**
  - **Type of relationship**: One-to-many (1:M), optional-to-mandatory

- **Description:** A vendor can have zero or many orders; every order is placed with exactly one vendor.
- **Minimum Cardinality**: A vendor may have 0 orders.
- **Maximum Cardinality**: Many orders per vendor.

- **Employee and Order**
  - **Type of relationship**: One-to-many (1:M), optional-to-mandatory
  - **Description:** An employee can place zero or many orders; every order is created by exactly one employee.
  - **Minimum Cardinali**ty: An employee may have 0 orders.
  - **Maximum Cardinality**: Many orders per employee.

- **Order and ORDER_ITEMS**
  - Type of relationship: One-to-many (1:M), optional-to-mandatory
  - **Description:** An order can contain zero or many line items; every line item belongs to exactly one order.
  - **Minimum Cardinality**: An order may have 0 line items (if drafts are allowed).
  - **Maximum Cardinality**: Many line items per order.

- **ORDER_ITEMS and PRIZE_INVENTORY**
  - **Type of relationship**: Many-to-one (M:1), mandatory-to-optional (equivalently, PRIZE_INVENTORY 1:M optional-to-mandatory)
  - **Description:** Each order line references exactly one catalog item; a catalog item can appear in zero or many order lines.
  - **Minimum Cardinality** (per item): A catalog item may appear in 0 order lines.
  - **Maximum Cardinality** (per item): Many order lines can reference the same item.

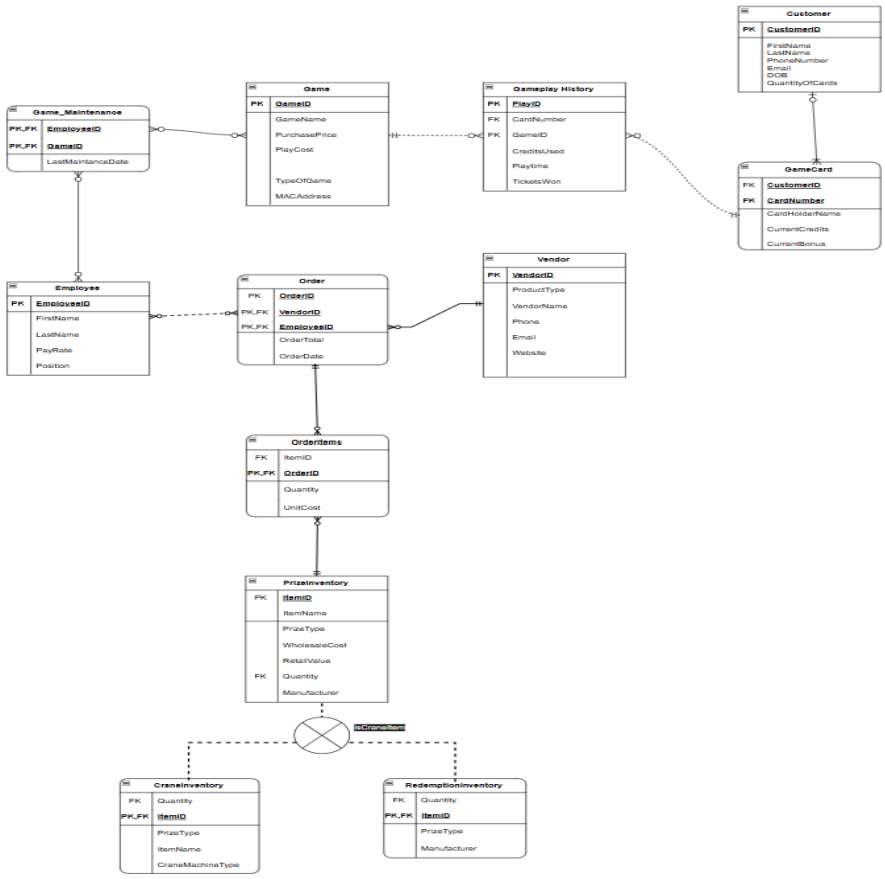
- **PRIZE_INVENTORY and CRANE_INVENTORY (ISA)**

- **Type of relationship**: IS-A. subtypes
- **Description:** Each crane-inventory row corresponds to exactly one total-inventory row with the same ItemID (inherited key).
- **Cardinality**: 1:1 between a PRIZE_INVENTORY record and its CRANE_INVENTORY record (when present).

- **PRIZE_INVENTORY and REDEMPTION_INVENTORY (ISA)**
  - **Type of relationship**: IS-A. Subtypes
  - **Description:** Each redemption-inventory row corresponds to exactly one total-inventory row with the same ItemID (inherited key).
  - **Cardinality**: 1:1 between a PRIZE_INVENTORY record and its REDEMPTION_INVENTORY record (when present).

3. Database Design

3.1 Transformed Database Design Diagram



3.2 List of Functional Dependencies

CUSTOMER

- CustomerID → FirstName, LastName, PhoneNumber, Email, DOB, QuantityOfCards

The CustomerID uniquely determines all customer attributes.

GAME_CARD

- CardNumber → CardHolderName, CurrentCredits, CurrentBonus, CustomerID

Each game card has one unique card number that determines ownership and balances.

GAME

- GameID → GameName, PurchasePrice, PlayCost, TypeOfGame, MACAddress

A game's unique ID determines all properties of the arcade machine.

GAMEPLAY_HISTORY

- PlayID → CardNumber, GameID, PlayTime, CreditsUsed, TicketsWon

Each play event is uniquely identified by PlayID and determines all related play details.

EMPLOYEE

- EmployeeID → FirstName, LastName, PayRate, Position

EmployeeID uniquely determines an employee's personal and compensation information.

VENDOR

- VendorID → VendorName, Phone, Email, Website, ProductType

VendorID uniquely determines all attributes of a vendor.

"ORDER"

- OrderID → OrderDate, OrderTotal, VendorID, EmployeeID

Each order has one unique identifier that determines its vendor, employee, and totals.

ORDER_ITEMS

Composite key dependency:

- (OrderID, ItemID) → Quantity, UnitCost

The combination of OrderID and ItemID uniquely identifies each order line entry.

PRIZE_INVENTORY

- ItemID → ItemName, PrizeType, Manufacturer, WholesaleCost, RetailValue, Quantity

Each prize inventory item is fully determined by its ItemID.

CRANE_INVENTORY

Subtype dependency (ISA relationship):

- ItemID → ItemName, PrizeType, Quantity, CraneMachineType

ItemID inherited from PRIZE_INVENTORY determines crane-specific attributes.

REDEMPTION_INVENTORY

Subtype dependency:

- ItemID → Quantity, TicketPrice

ItemID determines redemption pricing and stock.

GAME_MAINTENANCE

Composite key dependency:

- (GameID, EmployeeID) → LastMaintenanceDate

A maintenance record is uniquely identified by which employee performed work on which game.

4. Database Implementation

4.1 SQL Code - CREATE TABLE Scripts

Game:

```
CREATE TABLE GAME (
        GameID              Int             NOT NULL,
        GameName        Char(30)        NOT NULL,
        PurchasePrice     Int             NOT NULL,
        PlayCost            Int             NULL,
        TypeOfGame      Char(30)        NULL,
        MACAddress        Char(25)          NOT NULL,

        CONSTRAINT GamePK          PRIMARY KEY(GameID)
);
```

Game Maintenance:

```
CREATE TABLE Game_Maintenance (
            LastMaintenanceDate DATE    NOT NULL,
            GameID            NUMBER  NOT NULL,
            EmployeeID        NUMBER  NOT NULL,

            CONSTRAINT GameMainPK PRIMARY KEY (GameID, EmployeeID),

            CONSTRAINT GameMain_FK FOREIGN KEY (GameID)
               REFERENCES GAME (GameID)
               ON DELETE CASCADE,

            CONSTRAINT EmployeeMain_FK FOREIGN KEY (EmployeeID)
               REFERENCES EMPLOYEE (EmployeeID)
               ON DELETE CASCADE
        );
```

Employee:

```
CREATE TABLE EMPLOYEE (
        EmployeeID   Int            NOT NULL,
        FirstName    Char(20)       NOT NULL,
```

```
        LastName      Char(20)      NOT NULL,
        PayRate       DECIMAL       NOT NULL,
        Position      Char(30)      NULL,
        CONSTRAINT EmployeePK      PRIMARY KEY(EmployeeID),
        CONSTRAINT ValidPayRate CHECK
                              (PayRate >= 0)
);
```

Customer:

```
CREATE TABLE CUSTOMER (
   CustomerID      NUMBER        NOT NULL,
   FirstName       CHAR(25)      NOT NULL,
   LastName        CHAR(25)      NOT NULL,
   PhoneNumber     CHAR(14)      NULL,
   Email           VARCHAR2(50)  NOT NULL,
   DOB             DATE          NOT NULL,
   QuantityOfCards NUMBER        NOT NULL,

   CONSTRAINT CUSTOMERPK PRIMARY KEY (CustomerID),
   UNIQUE (FirstName, LastName),
   CONSTRAINT ValidBirthYear CHECK (
      EXTRACT(YEAR FROM DOB) BETWEEN 1950 AND 2100
   )
);
```

Game Card:

```
CREATE TABLE GAME_CARD (
   CardNumber     NUMBER        NOT NULL,
   CardHolderName VARCHAR2(30)  NOT NULL,
   CurrentCredits NUMBER        NOT NULL,
   CurrentBonus   NUMBER        NULL,
   CustomerID     NUMBER        NOT NULL,

   CONSTRAINT CardNumberPK PRIMARY KEY (CardNumber),
   CONSTRAINT CustomerIDFK FOREIGN KEY (CustomerID)
      REFERENCES CUSTOMER (CustomerID),
   CONSTRAINT ValidCredits CHECK (CurrentCredits >= 0)
);
```

Gameplay History:

```
CREATE TABLE GAMEPLAY_HISTORY (
   PlayID      NUMBER        NOT NULL,
   CardNumber  NUMBER        NOT NULL,
   GameID      NUMBER        NOT NULL,
   PlayTime    DATE          NOT NULL,
   CreditsUsed NUMBER(4)     NOT NULL,
   TicketsWon  NUMBER        NOT NULL,

   CONSTRAINT PlayIDPK PRIMARY KEY (PlayID),
   CONSTRAINT CardNumberFK FOREIGN KEY (CardNumber)
      REFERENCES GAME_CARD (CardNumber)
      ON DELETE CASCADE,
   CONSTRAINT GameIDFK FOREIGN KEY (GameID)
      REFERENCES GAME (GameID)
      ON DELETE CASCADE
);
```

Order:

```
CREATE TABLE "ORDER" (
   OrderID    NUMBER       NOT NULL,
   VendorID   NUMBER       NOT NULL,
   EmployeeID NUMBER       NOT NULL,
   OrderTotal NUMBER(10,2) NOT NULL,
   OrderDate  DATE         NOT NULL,

   CONSTRAINT OrderPK PRIMARY KEY (OrderID),
   CONSTRAINT OrderFK_Vendor FOREIGN KEY (VendorID)
      REFERENCES VENDOR (VendorID)
      ON DELETE CASCADE,
   CONSTRAINT OrderFK_Employee FOREIGN KEY (EmployeeID)
      REFERENCES EMPLOYEE (EmployeeID)
      ON DELETE CASCADE
);
```

Order Items:

```
CREATE TABLE ORDER_ITEMS (
   OrderID  NUMBER       NOT NULL,
   ItemID   NUMBER       NOT NULL,
   Quantity NUMBER(4)    NULL,
   UnitCost NUMBER(10,2) NULL,

   CONSTRAINT OrderItemsPK PRIMARY KEY (OrderID, ItemID),
   CONSTRAINT OrderItems_OrderFK FOREIGN KEY (OrderID)
      REFERENCES "ORDER" (OrderID)
      ON DELETE CASCADE,
   CONSTRAINT OrderItems_ItemFK FOREIGN KEY (ItemID)
      REFERENCES PRIZE_INVENTORY (ItemID)
      ON DELETE CASCADE
);
```

Vendor:

```
CREATE TABLE VENDOR (
   VendorID    NUMBER       NOT NULL,
   ProductType VARCHAR2(20) NOT NULL,
   VendorName  CHAR(25)     NOT NULL,
   Phone       CHAR(25)     NULL,
   Email       CHAR(25)     NULL,
   Website     CHAR(25)     NULL,

   CONSTRAINT VendorPK PRIMARY KEY (VendorID),
   UNIQUE (Email, Phone, Website),
   CONSTRAINT ProductTypeCK CHECK (
      ProductType IN ('Plush', 'Electronic', 'Miscellaneous')
   )
);
```

Prize_Inventory:

```
CREATE TABLE PRIZE_INVENTORY (
        ItemID              Int              NOT NULL,
        ItemName            Char(25)         NOT NULL,
        PrizeType           Char(25)         NULL,
        Manufacturer        Char(25)         NULL,
        WholesaleCost NUMBER(10,2)           NULL,
```

```
        RetailValue NUMBER(10,2)                NULL,
        Quantity                Numeric(4)   NULL,

CONSTRAINT        PRIZE_INVENTORYPK     PRIMARY KEY(ItemID)
);
```

Crane_Inventory:

```
CREATE TABLE CRANE_INVENTORY (
        ItemID                  Int             NOT NULL,
        ItemName                Char(25)        NOT NULL,
        PrizeType               Char(25)        NULL,
        Quantity                Numeric(4)      NULL,
        CraneMachineType Char(25)               NULL,

CONSTRAINT        CRANE_INVENTORYPK   PRIMARY KEY(ItemID),
CONSTRAINT        CRANE_INVENTORYFK1
        FOREIGN KEY (ItemID)
        REFERENCES   Prize_Inventory(ItemID)
);
```

Redemption_Inventory:

```
CREATE TABLE REDEMPTION_INVENTORY (
        ItemID                  Int             NOT NULL,
        Quantity                Numeric(4)      NULL,
        TicketPrice             Numeric(6)      NULL,

CONSTRAINT        REDEMPTION_INVENTORYPK  PRIMARY KEY(ItemID),
CONSTRAINT        REDEMPTION_INVENTORYFK1
        FOREIGN KEY (ItemID)
        REFERENCES   Prize_Inventory(ItemID)
);
```

4.2 SQL Code - INSERT Scripts

GAME:

```
INSERT INTO GAME (GameID, GameName, PurchasePrice, PlayCost, TypeOfGame,
MACAddress)

VALUES (1001, 'Dizzy Chicken', 9500, 2, NULL, '00-1B-44-11-3A-B7');

INSERT INTO GAME (GameID, GameName, PurchasePrice, PlayCost, TypeOfGame,
MACAddress)
VALUES (1002, 'Skeeball 1', 3000, 2, NULL, '00-2B-55-22-4A-C8');

INSERT INTO GAME (GameID, GameName, PurchasePrice, PlayCost, TypeOfGame,
MACAddress)
VALUES (1003, 'Skeeball 2', 3000, 2, NULL, '00-3A-12-44-5C-B9');

INSERT INTO GAME (GameID, GameName, PurchasePrice, PlayCost, TypeOfGame,
MACAddress)
VALUES (1004, 'E-Claw 1', 8000, 4, NULL, '00-4A-23-11-3A-B1');

INSERT INTO GAME (GameID, GameName, PurchasePrice, PlayCost, TypeOfGame,
MACAddress)
VALUES (1005, 'E-Claw 2', 8000, 8, NULL, '00-1A-22-33-4B-C5');

INSERT INTO GAME (GameID, GameName, PurchasePrice, PlayCost, TypeOfGame,
MACAddress)
VALUES (1006, 'E-Claw Cosmic', 20000, 20, NULL, '00-2B-33-91-2A-B6');

INSERT INTO GAME (GameID, GameName, PurchasePrice, PlayCost, TypeOfGame,
MACAddress)
VALUES (1007, 'Monopoly Roll-N-Go', 12000, 4, NULL, '00-2A-99-45-3B-A1');

INSERT INTO GAME (GameID, GameName, PurchasePrice, PlayCost, TypeOfGame,
MACAddress)
VALUES (1008, 'Sink It', 10000, 2, NULL, '00-6B-10-23-3A-C2');

INSERT INTO GAME (GameID, GameName, PurchasePrice, PlayCost, TypeOfGame,
MACAddress)
VALUES (1009, 'Harpoon Lagoon', 22000, 4, NULL, '00-7A-33-21-4C-B1');

INSERT INTO GAME (GameID, GameName, PurchasePrice, PlayCost, TypeOfGame,
MACAddress)
VALUES (1010, 'Space Warp', 6000, 4, NULL, '00-2B-67-23-7A-C4');
```

_____

GAMEPLAY_HISTORY

```
INSERT INTO GAMEPLAY_HISTORY
(PlayID, CardNumber, GameID, PlayTime, CreditsUsed, TicketsWon)
VALUES
(1, 11234, 1001, TO_DATE('2025-06-20 14:15:00','YYYY-MM-DD HH24:MI:SS'), 2, 45);

INSERT INTO GAMEPLAY_HISTORY
(PlayID, CardNumber, GameID, PlayTime, CreditsUsed, TicketsWon)
VALUES
(2, 11235, 1002, TO_DATE('2025-06-20 14:18:30','YYYY-MM-DD HH24:MI:SS'), 2, 10);

INSERT INTO GAMEPLAY_HISTORY
(PlayID, CardNumber, GameID, PlayTime, CreditsUsed, TicketsWon)
VALUES
(3, 11236, 1009, TO_DATE('2025-06-21 16:05:12','YYYY-MM-DD HH24:MI:SS'), 4,
120);

INSERT INTO GAMEPLAY_HISTORY
(PlayID, CardNumber, GameID, PlayTime, CreditsUsed, TicketsWon)
VALUES
(4, 11237, 1008, TO_DATE('2025-06-21 16:10:55','YYYY-MM-DD HH24:MI:SS'), 4, 60);

INSERT INTO GAMEPLAY_HISTORY
(PlayID, CardNumber, GameID, PlayTime, CreditsUsed, TicketsWon)
VALUES
(5, 11238, 1004, TO_DATE('2025-07-01 13:02:08','YYYY-MM-DD HH24:MI:SS'), 4, 30);

INSERT INTO GAMEPLAY_HISTORY
(PlayID, CardNumber, GameID, PlayTime, CreditsUsed, TicketsWon)
VALUES
(6, 11239, 1005, TO_DATE('2025-07-01 13:05:40','YYYY-MM-DD HH24:MI:SS'), 8, 95);

INSERT INTO GAMEPLAY_HISTORY
(PlayID, CardNumber, GameID, PlayTime, CreditsUsed, TicketsWon)
VALUES
(7, 11234, 1003, TO_DATE('2025-07-04 18:22:10','YYYY-MM-DD HH24:MI:SS'), 2, 25);

INSERT INTO GAMEPLAY_HISTORY
```

(PlayID, CardNumber, GameID, PlayTime, CreditsUsed, TicketsWon)
VALUES
(8, 11235, 1010, TO_DATE('2025-07-08 19:11:47','YYYY-MM-DD HH24:MI:SS'), 4, 80);

CUSTOMER:

```
INSERT INTO CUSTOMER
(CustomerID, FirstName, LastName, PhoneNumber, Email, DOB,
QuantityOfCards)
VALUES (1000, 'John', 'Jones', '555-5551', 'jjones@gmail.com',
    TO_DATE('2005-04-01','YYYY-MM-DD'), 2);

INSERT INTO CUSTOMER
(CustomerID, FirstName, LastName, PhoneNumber, Email, DOB,
QuantityOfCards)
VALUES (1001, 'Sarah', 'Miller', '555-5552', 'smiller@gmail.com',
    TO_DATE('2003-08-12','YYYY-MM-DD'), 1);

INSERT INTO CUSTOMER
(CustomerID, FirstName, LastName, PhoneNumber, Email, DOB,
QuantityOfCards)
VALUES (1002, 'Michael', 'Brown', '555-5553', 'mbrown@yahoo.com',
    TO_DATE('1999-11-23','YYYY-MM-DD'), 3);

INSERT INTO CUSTOMER
(CustomerID, FirstName, LastName, PhoneNumber, Email, DOB,
QuantityOfCards)
VALUES (1003, 'Emily', 'Davis', '555-5554', 'edavis@hotmail.com',
    TO_DATE('2001-02-17','YYYY-MM-DD'), 1);

INSERT INTO CUSTOMER
(CustomerID, FirstName, LastName, PhoneNumber, Email, DOB,
QuantityOfCards)
VALUES (1004, 'Daniel', 'Lopez', '555-5555', 'dlopez@gmail.com',
    TO_DATE('2004-06-05','YYYY-MM-DD'), 4);

INSERT INTO CUSTOMER
(CustomerID, FirstName, LastName, PhoneNumber, Email, DOB,
QuantityOfCards)
VALUES (1005, 'Ava', 'Wilson', '555-5556', 'awilson@outlook.com',
```

TO_DATE('2002-10-30','YYYY-MM-DD'), 2);

GAMECARD:

INSERT INTO GAME_CARD (CustomerID, CardNumber, CardHolderName, CurrentCredits, CurrentBonus)
VALUES (1000, 11234, 'jasonh', 15230, 12);

INSERT INTO GAME_CARD (CustomerID, CardNumber, CardHolderName, CurrentCredits, CurrentBonus)
VALUES (1001, 11235, 'sarahm', 8340, 5);

INSERT INTO GAME_CARD (CustomerID, CardNumber, CardHolderName, CurrentCredits, CurrentBonus)
VALUES (1002, 11236, 'mikeb', 4500, 8);

INSERT INTO GAME_CARD (CustomerID, CardNumber, CardHolderName, CurrentCredits, CurrentBonus)
VALUES (1003, 11237, 'emilyd', 12000, 3);

INSERT INTO GAME_CARD (CustomerID, CardNumber, CardHolderName, CurrentCredits, CurrentBonus)
VALUES (1004, 11238, 'danlo', 7600, 10);

INSERT INTO GAME_CARD (CustomerID, CardNumber, CardHolderName, CurrentCredits, CurrentBonus)
VALUES (1005, 11239, 'avaw', 5300, 6);

VENDOR:

INSERT INTO VENDOR
(VendorID, ProductType, VendorName, Phone, Email, Website)
VALUES
(1, 'Plush', 'Plush World', '555-2312', 'pworld@hotmail.com', 'www.plushworld.com');

INSERT INTO VENDOR
(VendorID, ProductType, VendorName, Phone, Email, Website)
VALUES

```
        (2, 'Electronic', 'Gadget Galaxy', '555-7842', 'contact@gadgetgalaxy.com',
        'www.gadgetgalaxy.com');

INSERT INTO VENDOR
        (VendorID, ProductType, VendorName, Phone, Email, Website)
        VALUES
        (3, 'Miscellaneous', 'Arcade Supplies Co', '555-9931', 'info@arcadesupplies.co',
        'www.arcadesupplies.co');

INSERT INTO VENDOR
        (VendorID, ProductType, VendorName, Phone, Email, Website)
        VALUES
        (4, 'Plush', 'Soft Friends Inc', '555-6620', 'sales@softfriends.com',
'www.softfriends.com');

INSERT INTO VENDOR
        (VendorID, ProductType, VendorName, Phone, Email, Website)
        VALUES
        (5, 'Electronic', 'TechFun Motors', '555-4129', 'support@techfunmotors.net',
        'www.techfunmotors.net');

INSERT INTO VENDOR
        (VendorID, ProductType, VendorName, Phone, Email, Website)
        VALUES
        (6, 'Miscellaneous', 'Prize Depot', '555-7784', 'orders@prizedepot.com',
        'www.prizedepot.com');

INSERT INTO VENDOR
        (VendorID, ProductType, VendorName, Phone, Email, Website)
        VALUES
        (7, 'Plush', 'CuddleWorks', '555-8085', 'hello@cuddleworks.org',
'www.cuddleworks.org');
```

GAME_MAINTENANCE:

```
INSERT INTO Game_Maintenance (GameID, EmployeeID, LastMaintenanceDate)
VALUES (1002, 100, TO_DATE('2025-06-20','YYYY-MM-DD'));

INSERT INTO Game_Maintenance (GameID, EmployeeID, LastMaintenanceDate)
VALUES (1009, 101, TO_DATE('2025-06-21','YYYY-MM-DD'));
```

```
INSERT INTO Game_Maintenance (GameID, EmployeeID, LastMaintenanceDate)
VALUES (1006, 100, TO_DATE('2025-06-30','YYYY-MM-DD'));

INSERT INTO Game_Maintenance (GameID, EmployeeID, LastMaintenanceDate)
VALUES (1007, 109, TO_DATE('2025-07-01','YYYY-MM-DD'));

INSERT INTO Game_Maintenance (GameID, EmployeeID, LastMaintenanceDate)
VALUES (1002, 101, TO_DATE('2025-07-04','YYYY-MM-DD'));

INSERT INTO Game_Maintenance (GameID, EmployeeID, LastMaintenanceDate)
VALUES (1005, 100, TO_DATE('2025-07-08','YYYY-MM-DD'));

INSERT INTO Game_Maintenance (GameID, EmployeeID, LastMaintenanceDate)
VALUES (1001, 104, TO_DATE('2025-07-12','YYYY-MM-DD'));

INSERT INTO Game_Maintenance (GameID, EmployeeID, LastMaintenanceDate)
VALUES (1010, 109, TO_DATE('2025-07-28','YYYY-MM-DD'));

INSERT INTO Game_Maintenance (GameID, EmployeeID, LastMaintenanceDate)
VALUES (1003, 100, TO_DATE('2025-08-04','YYYY-MM-DD'));

INSERT INTO Game_Maintenance (GameID, EmployeeID, LastMaintenanceDate)
VALUES (1004, 101, TO_DATE('2025-08-16','YYYY-MM-DD'));
```

—----------

EMPLOYEE

```
INSERT INTO EMPLOYEE (EmployeeID, FirstName, LastName, PayRate, Position)
VALUES (100, 'Sean', 'Collins', 1000000, 'Senior Manager');

INSERT INTO EMPLOYEE (EmployeeID, FirstName, LastName, PayRate, Position)
VALUES (101, 'Aria', 'Lindberg', 23, 'Senior Manager');

INSERT INTO EMPLOYEE (EmployeeID, FirstName, LastName, PayRate, Position)
VALUES (102, 'Robert', 'Lydon', 14, 'Customer Service Representative');
```

```
INSERT INTO EMPLOYEE (EmployeeID, FirstName, LastName, PayRate, Position)
VALUES (103, 'Jessica', 'Lydon', 13, 'Customer Service Representative');

INSERT INTO EMPLOYEE (EmployeeID, FirstName, LastName, PayRate, Position)
VALUES (104, 'Natalie', 'Burrell', 21, 'Manager');

INSERT INTO EMPLOYEE (EmployeeID, FirstName, LastName, PayRate, Position)
VALUES (105, 'Allie', 'Desiderio', 20, 'Manager');

INSERT INTO EMPLOYEE (EmployeeID, FirstName, LastName, PayRate, Position)
VALUES (106, 'Bob', 'Millar', 20, 'Manager');

INSERT INTO EMPLOYEE (EmployeeID, FirstName, LastName, PayRate, Position)
VALUES (107, 'Aidan', 'Millar', 13, 'Customer Service Representative');

INSERT INTO EMPLOYEE (EmployeeID, FirstName, LastName, PayRate, Position)
VALUES (108, 'Collette', 'Millar', 14, 'Customer Service Representative');

INSERT INTO EMPLOYEE (EmployeeID, FirstName, LastName, PayRate, Position)
VALUES (109, 'Nathan', 'Collins', 1, 'Mascot');
```

ORDER

```
INSERT INTO "ORDER"
(OrderID, OrderDate, OrderTotal, VendorID, EmployeeID)
VALUES
(3001, TO_DATE('2025-05-15','YYYY-MM-DD'), 132.50, 1, 104);

INSERT INTO "ORDER"
(OrderID, OrderDate, OrderTotal, VendorID, EmployeeID)
VALUES
(3002, TO_DATE('2025-05-20','YYYY-MM-DD'), 210.00, 2, 105);

INSERT INTO "ORDER"
(OrderID, OrderDate, OrderTotal, VendorID, EmployeeID)
VALUES
(3003, TO_DATE('2025-05-25','YYYY-MM-DD'), 130.00, 6, 106);

INSERT INTO "ORDER"
(OrderID, OrderDate, OrderTotal, VendorID, EmployeeID)
```

```
VALUES
(3004, TO_DATE('2025-06-01','YYYY-MM-DD'), 175.00, 3, 102);

INSERT INTO "ORDER"
(OrderID, OrderDate, OrderTotal, VendorID, EmployeeID)
VALUES
(3005, TO_DATE('2025-06-05','YYYY-MM-DD'),  90.00, 4, 103);
```

ORDER ITEMS

```
INSERT INTO ORDER_ITEMS
(OrderID, ItemID, Quantity, UnitCost)
VALUES
(3001, 2001, 50, 1.25);

INSERT INTO ORDER_ITEMS
(OrderID, ItemID, Quantity, UnitCost)
VALUES
(3001, 2002, 20, 3.50);

INSERT INTO ORDER_ITEMS
(OrderID, ItemID, Quantity, UnitCost)
VALUES
(3002, 2003, 10, 15.00);

INSERT INTO ORDER_ITEMS
(OrderID, ItemID, Quantity, UnitCost)
VALUES
(3002, 2004, 30, 2.00);

INSERT INTO ORDER_ITEMS
(OrderID, ItemID, Quantity, UnitCost)
VALUES
(3003, 2005, 100, 0.40);

INSERT INTO ORDER_ITEMS
(OrderID, ItemID, Quantity, UnitCost)
VALUES
(3003, 2006, 5, 18.00);
```

PRIZE INVENTORY

```
INSERT INTO PRIZE_INVENTORY
(ItemID, ItemName, PrizeType, Manufacturer, WholesaleCost, RetailValue, Quantity)
VALUES
(2001, 'Small Plush Bear', 'Plush', 'Plush World', 1.25, 5.00, 150);

INSERT INTO PRIZE_INVENTORY
(ItemID, ItemName, PrizeType, Manufacturer, WholesaleCost, RetailValue, Quantity)
VALUES
(2002, 'Large Plush Dragon', 'Plush', 'CuddleWorks', 3.50, 12.00, 60);

INSERT INTO PRIZE_INVENTORY
(ItemID, ItemName, PrizeType, Manufacturer, WholesaleCost, RetailValue, Quantity)
VALUES
(2003, 'LED Headphones', 'Electronic', 'Gadget Galaxy', 15.00, 35.00, 40);

INSERT INTO PRIZE_INVENTORY
(ItemID, ItemName, PrizeType, Manufacturer, WholesaleCost, RetailValue, Quantity)
VALUES
(2004, 'Die-Cast Race Car', 'Toy', 'Arcade Supplies Co', 2.00, 7.50, 120);

INSERT INTO PRIZE_INVENTORY
(ItemID, ItemName, PrizeType, Manufacturer, WholesaleCost, RetailValue, Quantity)
VALUES
(2005, 'Candy Mix Bag', 'Confection', 'Prize Depot', 0.40, 2.00, 300);

INSERT INTO PRIZE_INVENTORY
(ItemID, ItemName, PrizeType, Manufacturer, WholesaleCost, RetailValue, Quantity)
VALUES
(2006, 'Bluetooth Speaker', 'Electronic', 'TechFun Motors', 18.00, 45.00, 25);
```

CRANE_INVENTORY

```
INSERT INTO CRANE_INVENTORY
(ItemID, ItemName, PrizeType, Quantity, CraneMachineType)
VALUES
(2001, 'Small Plush Bear', 'Plush', 80, 'E-Claw Standard');
```

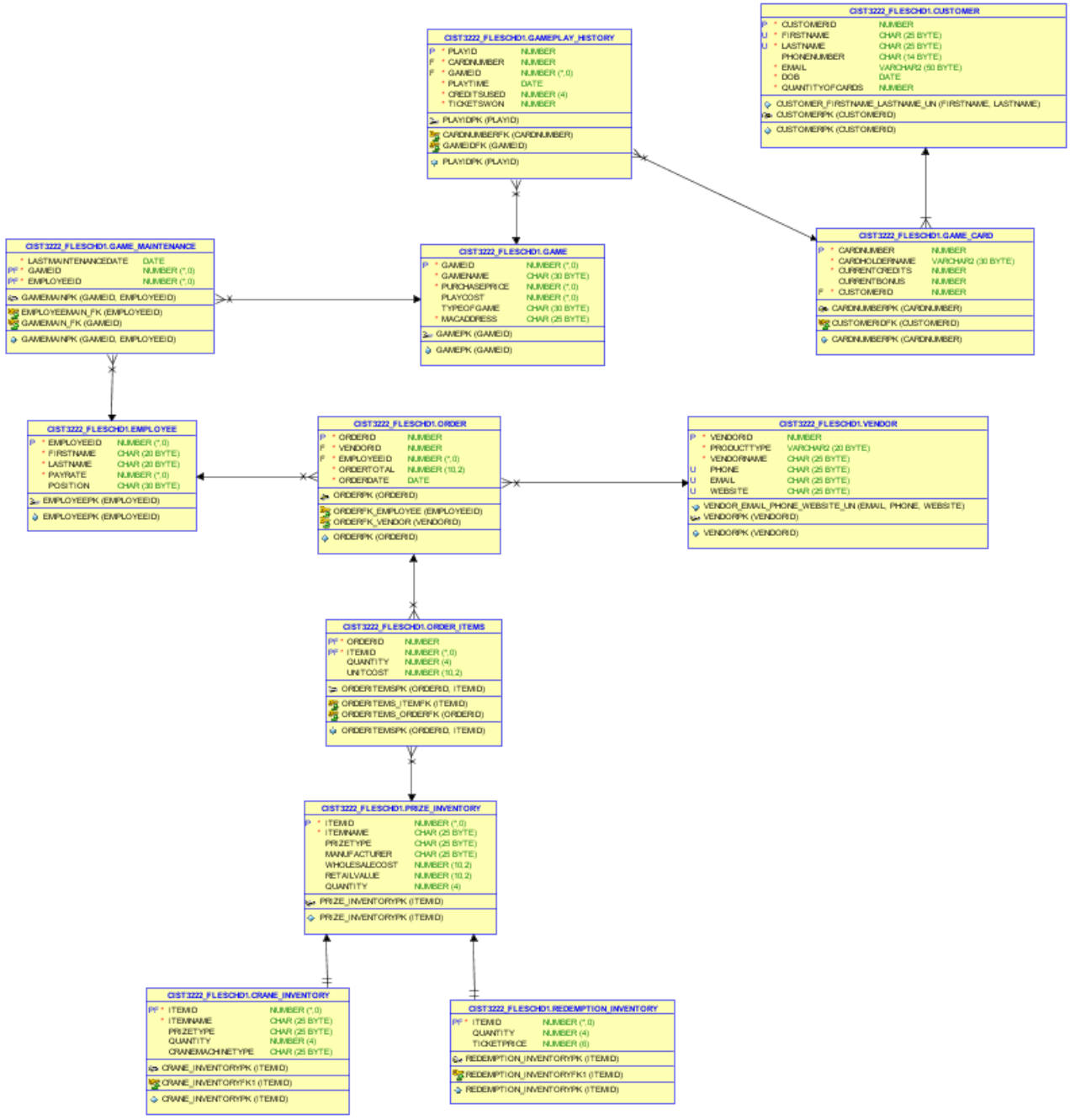
```
INSERT INTO CRANE_INVENTORY
(ItemID, ItemName, PrizeType, Quantity, CraneMachineType)
VALUES
(2002, 'Large Plush Dragon', 'Plush', 30, 'E-Claw Cosmic');

INSERT INTO CRANE_INVENTORY
(ItemID, ItemName, PrizeType, Quantity, CraneMachineType)
VALUES
(2004, 'Die-Cast Race Car', 'Toy', 60, 'Harpoon Lagoon Crane');

INSERT INTO CRANE_INVENTORY
(ItemID, ItemName, PrizeType, Quantity, CraneMachineType)
VALUES
(2005, 'Candy Mix Bag', 'Confection', 90, 'E-Claw Standard');

INSERT INTO CRANE_INVENTORY
(ItemID, ItemName, PrizeType, Quantity, CraneMachineType)
VALUES
(2003, 'LED Headphones', 'Electronic', 20, 'Cosmic Crane Premium');
```

REDEMPTION_INVENTORY

```
INSERT INTO REDEMPTION_INVENTORY
(ItemID, Quantity, TicketPrice)
VALUES
(2001, 70, 250);

INSERT INTO REDEMPTION_INVENTORY
(ItemID, Quantity, TicketPrice)
VALUES
(2003, 40, 1200);

INSERT INTO REDEMPTION_INVENTORY
(ItemID, Quantity, TicketPrice)
VALUES
(2005, 250, 80);

INSERT INTO REDEMPTION_INVENTORY
(ItemID, Quantity, TicketPrice)
VALUES
```

```
(2006, 20, 1800);

INSERT INTO REDEMPTION_INVENTORY
(ItemID, Quantity, TicketPrice)
VALUES
(2002, 45, 900);
```

4.3 Screenshots of ER Diagrams in SQL Developer



4.4 Screenshots of ALL Tables with Data

VENDOR

	VENDORID	PRODUCTTYPE	VENDORNAME	PHONE	EMAIL	WEBSITE
1	1	Plush	Plush World	555-2312	pworld@hotmail.com	www.plushworld.com
2	2	Electronic	Gadget Galaxy	555-7842	contact@gadgetgalaxy.com	www.gadgetgalaxy.com
3	3	Miscellaneous	Arcade Supplies Co	555-9931	info@arcadesupplies.co	www.arcadesupplies.co
4	4	Plush	Soft Friends Inc	555-6620	sales@softfriends.com	www.softfriends.com
5	5	Electronic	TechFun Motors	555-4129	support@techfunmotors.net	www.techfunmotors.net
6	6	Miscellaneous	Prize Depot	555-7784	orders@prizedepot.com	www.prizedepot.com
7	7	Plush	CuddleWorks	555-8085	hello@cuddleworks.org	www.cuddleworks.org

EMPLOYEE

	EMPLOYEEID	FIRSTNAME	LASTNAME	PAYRATE	POSITION
1	100	Sean	Collins	1000000	Senior Manager
2	101	Aria	Lindberg	23	Senior Manager
3	102	Robert	Lydon	14	Customer Service Representative
4	103	Jessica	Lydon	13	Customer Service Representative
5	104	Natalie	Burrell	21	Manager
6	105	Allie	Desiderio	20	Manager
7	106	Bob	Millar	20	Manager
8	107	Aidan	Millar	13	Customer Service Representative
9	108	Collette	Millar	14	Customer Service Representative
10	109	Nathan	Collins	1	Mascot

GAME

	GAMEID	GAMENAME	PURCHASEPRICE	PLAYCOST	TYPEOFGAME	MACADDRESS
1	1001	Dizzy Chicken	9500	2	Arcade	00-1B-44-11-3A-B7
2	1002	Skeeball 1	3000	2	Skeeball	00-2B-55-22-4A-C8
3	1003	Skeeball 2	3000	2	Skeeball	00-3A-12-44-5C-B9
4	1004	E-Claw 1	8000	4	Claw	00-4A-23-11-3A-B1
5	1005	E-Claw 2	8000	8	Claw	00-1A-22-33-4B-C5
6	1006	E-Claw Cosmic	20000	20	Claw	00-2B-33-91-2A-B6
7	1007	Monopoly Roll-N-Go	12000	4	Arcade	00-2A-99-45-3B-A1
8	1008	Sink It	10000	2	Arcade	00-6B-10-23-3A-C2
9	1009	Harpoon Lagoon	22000	4	Arcade	00-7A-33-21-4C-B1
10	1010	Space Warp	6000	4	Arcade	00-2B-67-23-7A-C4

CUSTOMER

	CUSTOMERID	FIRSTNAME	LASTNAME	PHONENUMBER	EMAIL	DOB	QUANTITYOFCARDS
1	1000	John	Jones	555-5551	jjones@gmail.com	01-APR-05	2
2	1001	Sarah	Miller	555-5552	smiller@gmail.com	12-AUG-03	1
3	1002	Michael	Brown	555-5553	mbrown@yahoo.com	23-NOV-99	3
4	1003	Emily	Davis	555-5554	edavis@hotmail.com	17-FEB-01	1
5	1004	Daniel	Lopez	555-5555	dlopez@gmail.com	05-JUN-04	4
6	1005	Ava	Wilson	555-5556	awilson@outlook.com	30-OCT-02	2

PRIZE_INVENTORY

	ITEMID	ITEMNAME	PRIZETYPE	MANUFACTURER	WHOLESALECOST	RETAILVALUE	QUANTITY
1	2001	Small Plush Bear	Plush	Plush World	1.25	5	150
2	2002	Large Plush Dragon	Plush	CuddleWorks	3.5	12	60
3	2003	LED Headphones	Electronic	Gadget Galaxy	15	35	40
4	2004	Die-Cast Race Car	Toy	Arcade Supplies Co	2	7.5	120
5	2005	Candy Mix Bag	Confection	Prize Depot	0.4	2	300
6	2006	Bluetooth Speaker	Electronic	TechFun Motors	18	45	25

GAME_CARD

	CARDNUMBER	CARDHOLDERNAME	CURRENTCREDITS	CURRENTBONUS	CUSTOMERID
1	11234	jasonh	15230	12	1000
2	11235	sarahm	8340	5	1001
3	11236	mikeb	4500	8	1002
4	11237	emilyd	12000	3	1003
5	11238	danlo	7600	10	1004
6	11239	avaw	5300	6	1005

GAMEPLAY_HISTORY

	PLAYID	CARDNUMBER	GAMEID	PLAYTIME	CREDITSUSED	TICKETSWON
1	1	11234	1001	20-JUN-25	2	45
2	2	11235	1002	20-JUN-25	2	10
3	3	11236	1009	21-JUN-25	4	120
4	4	11237	1008	21-JUN-25	4	60
5	5	11238	1004	01-JUL-25	4	30
6	6	11239	1005	01-JUL-25	8	95
7	7	11234	1003	04-JUL-25	2	25
8	8	11235	1010	08-JUL-25	4	80

"ORDER"

	ORDERID	VENDORID	EMPLOYEEID	ORDERTOTAL	ORDERDATE
1	3001	1	104	132.5	15-MAY-25
2	3002	2	105	210	20-MAY-25
3	3003	6	106	130	25-MAY-25
4	3004	3	102	175	01-JUN-25
5	3005	4	103	90	05-JUN-25

ORDER_ITEMS

	ORDERID	ITEMID	QUANTITY	UNITCOST
1	3001	2001	50	1.25
2	3001	2002	20	3.5
3	3002	2003	10	15
4	3002	2004	30	2
5	3003	2005	100	0.4
6	3003	2006	5	18

GAME_MAINTENANCE

	LASTMAINTENANCEDATE	GAMEID	EMPLOYEEID
1	20-JUN-25	1002	100
2	21-JUN-25	1009	101
3	30-JUN-25	1006	100
4	01-JUL-25	1007	109
5	04-JUL-25	1002	101
6	08-JUL-25	1005	100
7	12-JUL-25	1001	104
8	28-JUL-25	1010	109
9	04-AUG-25	1003	100
10	16-AUG-25	1004	101

CRANE_INVENTORY

	ITEMID	ITEMNAME	PRIZETYPE	QUANTITY	CRANEMACHINETYPE
1	2001	Small Plush Bear	Plush	80	E-Claw Standard
2	2002	Large Plush Dragon	Plush	30	E-Claw Cosmic
3	2004	Die-Cast Race Car	Toy	60	Harpoon Lagoon Crane
4	2005	Candy Mix Bag	Confection	90	E-Claw Standard
5	2003	LED Headphones	Electronic	20	Cosmic Crane Premium

REDEMPTION_INVENTORY

	ITEMID	QUANTITY	TICKETPRICE
1	2001	70	250
2	2003	40	1200
3	2005	250	80
4	2006	20	1800
5	2002	45	900